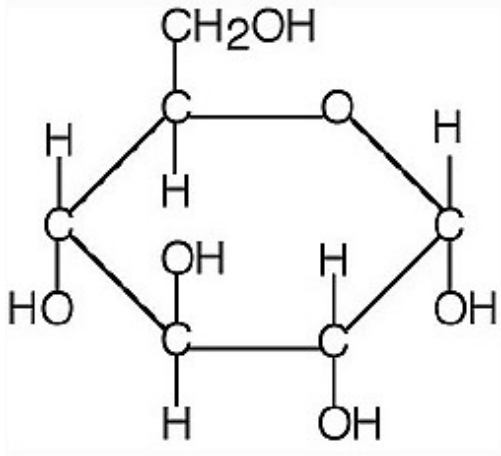
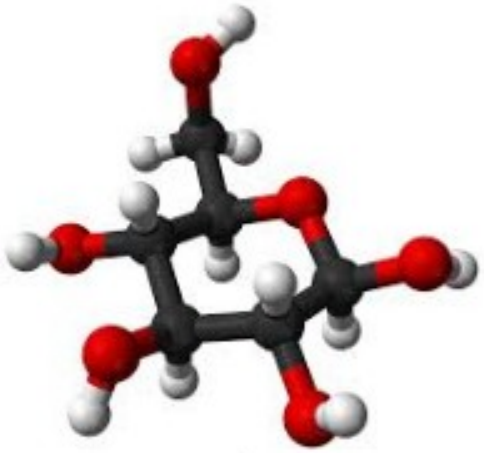
Blood Sugar Homeostasis



Glucose

- Sugar is transported in the blood as **Glucose**
- Glucose ($C_6H_{12}O_6$)
  - Monosaccharide (simple sugar)
  - Source of energy for cell's activities
  - Digested carbohydrate
  - Enters blood stream through intestinal wall
  - Transported in blood – enters body cells
  - Energy released during cellular respiration in mitochondria





$$C_6H_{12}O_6 + 6O_2 \Rightarrow 6CO_2 + 6H_2O + ATP$$

Glucose Oxygen Carbon Dioxide Water Energy

- Excess glucose stored as glycogen in liver and muscles, also converted to fat.

Learning Aim

Blood Glucose

- Stable glucose levels in blood mean stable supply to cells.
- Normal range: 3.9-5.5 mmol/L (70-140 mg/100mL)
- Rises sharply after a meal
- Homeostatic mechanisms operate to bring it down to normal levels.
- Excess glucose removed and stored ready for release back into the bloodstream if levels fall below normal range:
    - As glycogen in liver and muscles
    - Glycogen – long chains of glucose

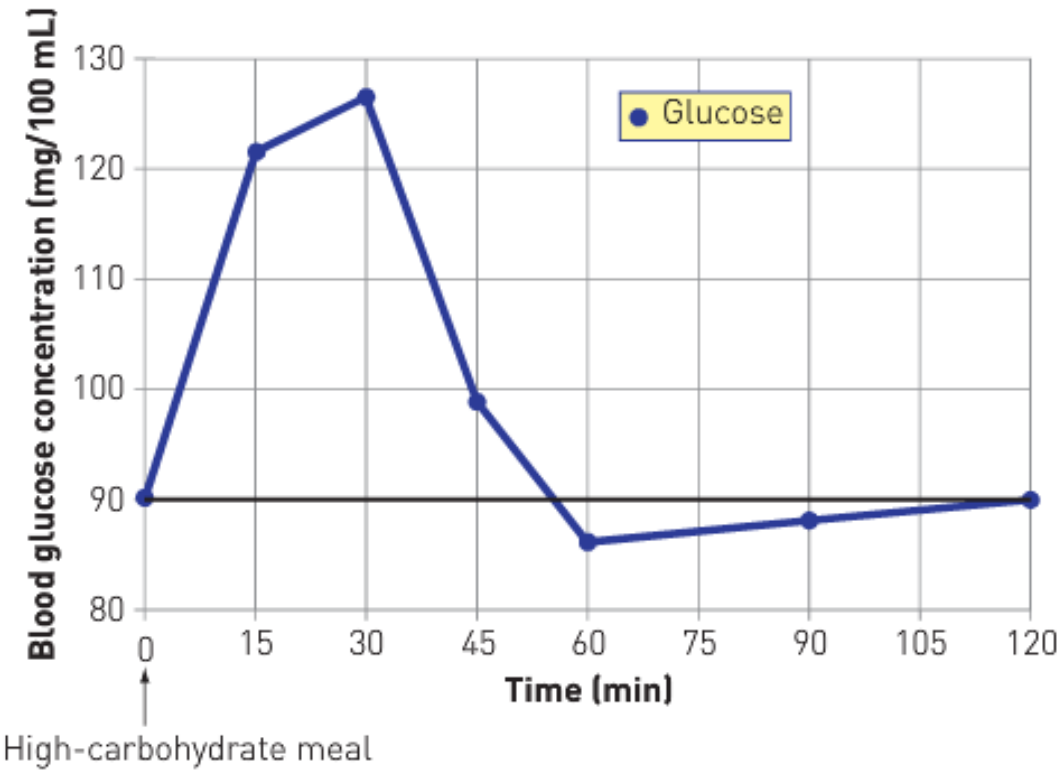


Figure 8.2 Change in blood glucose concentration over time following a high-carbohydrate meal

Learning Aim: Describe the normal range for blood g

Regulation of Blood Glucose

- Under hormonal control

- Pancreas:
  - Insulin
  - Glucagon

- Adrenal glands
  - Cortisol
  - Adrenalin and Noradrenalin

- Liver is involved in storage and release of glucose in response to hormones

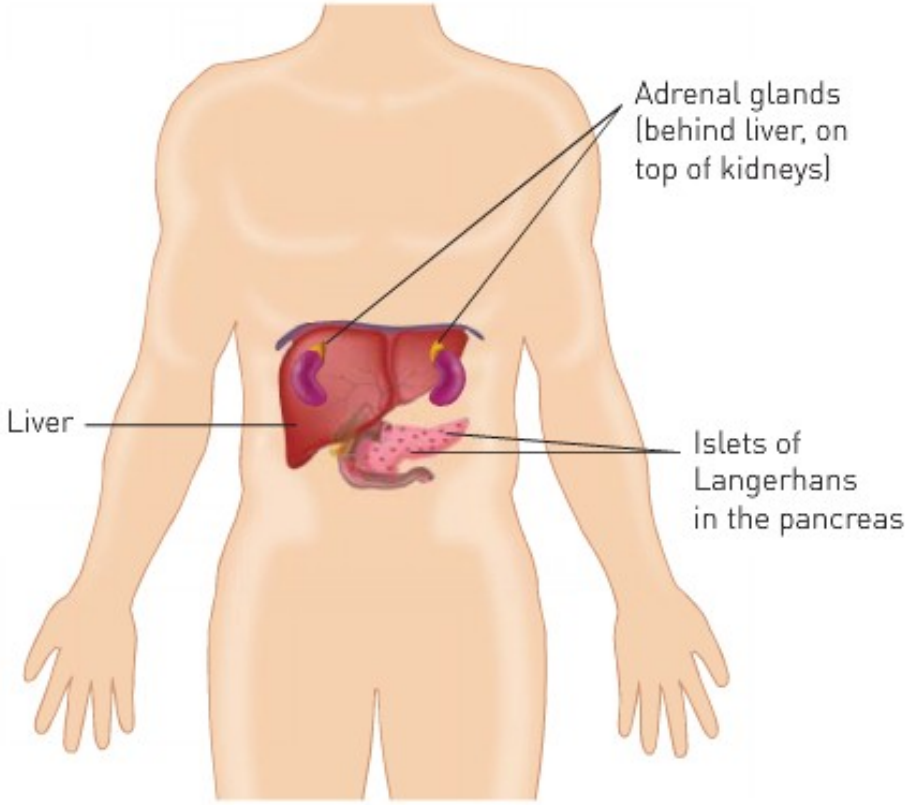


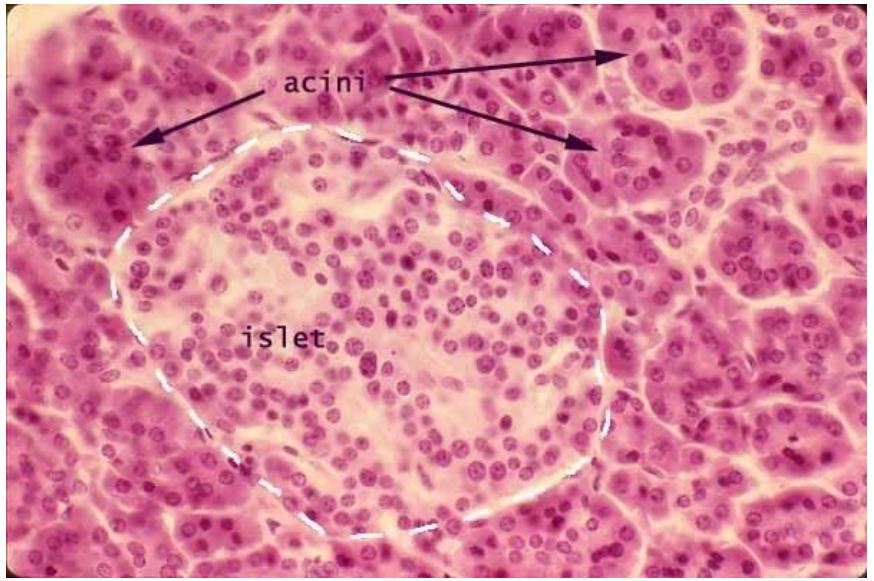
Figure 8.3 Main organs involved in the regulation of blood sugar

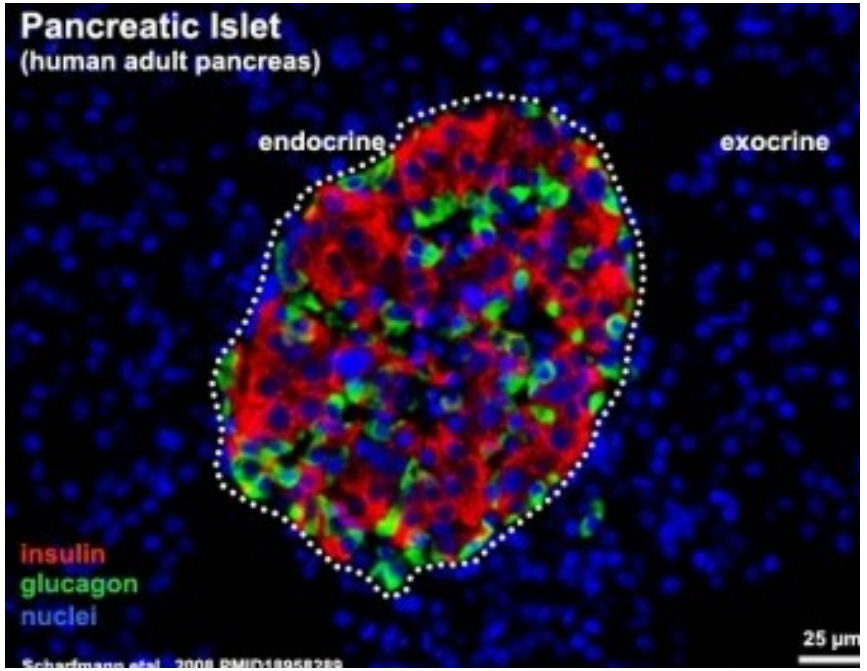
Learning Aim: List the main organs involv

Role of the pancreas

Pancreas:

- Contains clusters of insulin secreting cells “Islets of Langerhans” or “Pancreatic Islets”
  - Alpha cells
    - Detect blood glucose levels (*receptor)*
    - Act to raise blood glucose when it is too low (*modulator)* by secreting glucagon *(messenger)*.
  - Beta cells
    - Detect blood glucose levels (*receptor)*
    - Act to lower blood glucose when it is too high (*modulator)* by secreting insulin (*messenger)*





Learning Aim: Dis

Role of Liver in blood glucose homeostasis

- Liver is an *effector* for blood glucose homeostasis (along with body cells)
  - In upper abdomen
  - Able to convert
    - glucose → glycogen (storage): *glycogenesis*
    - glycogen → glucose (release back into blood): *glycogenolysis*
    - protein → glucose (release into blood) *gluconeogenesis*
  - Hepatic portal vein
    - Brings blood to liver from digestive tract
      - Glucose may be removed for storage as glycogen
      - May be used to provide energy for liver function
      - May remain in blood to go to other cells
    - Excess blood glucose (after glycogen) converted to fat for long term storage (*lipogenesis)*. Lipogenesis occurs in the liver and fat cells.

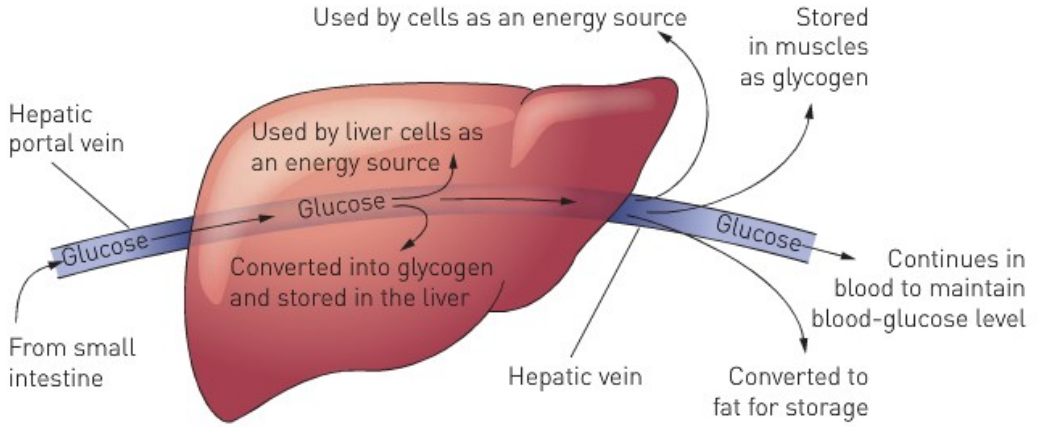


Figure 8.5 The fate of glucose absorbed in the small intestine

Liver – glycogen storage

- Body can store 500g of glycogen:
  - 100g in liver
  - 400g in skeletal muscle cells – used to power muscle contraction
- Glycogen:
  - Formed by combining glucose molecules into long chains
  - "glycogenesis" – in liver – stimulated by ***insulin*** from pancreas
  - Can't be used directly by cells: must be converted back to glucose
  - "glycogenolysis" – in liver/muscles – stimulated by ***glucagon*** from pancreas

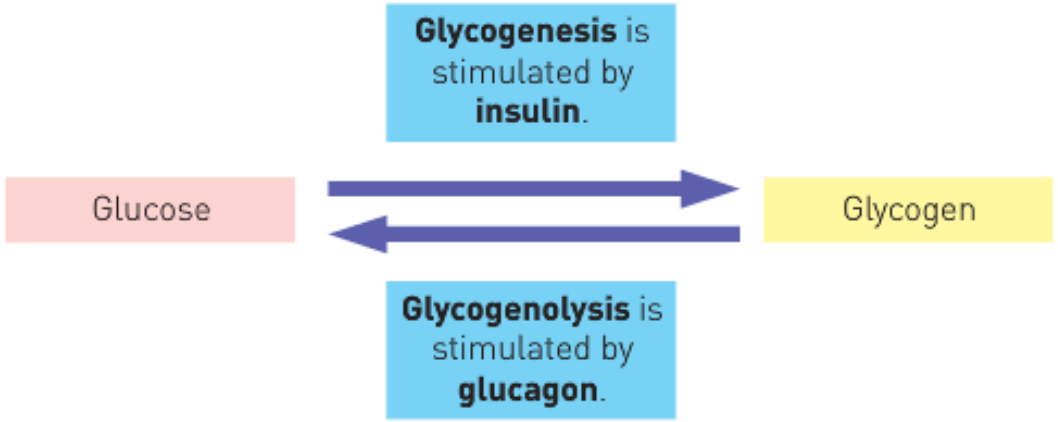
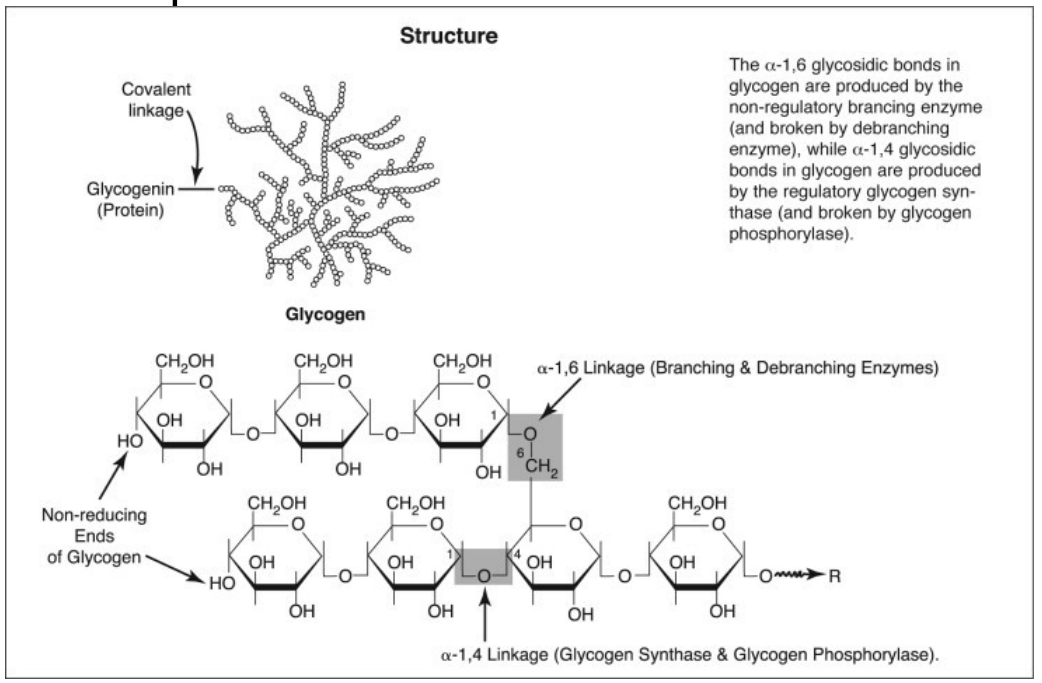


Figure 8.6 Glucose–glycogen conversions stimulated by pancreatic hormones



Responses to Insulin and Glucagon

Insulin Effects

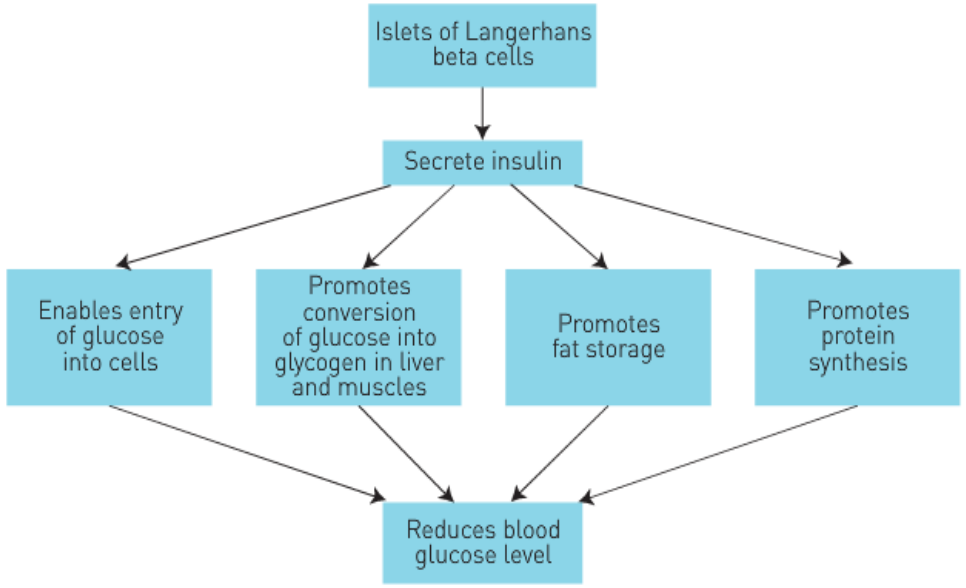


Figure 8.7 Effects of insulin on blood glucose levels

Glucagon Effects

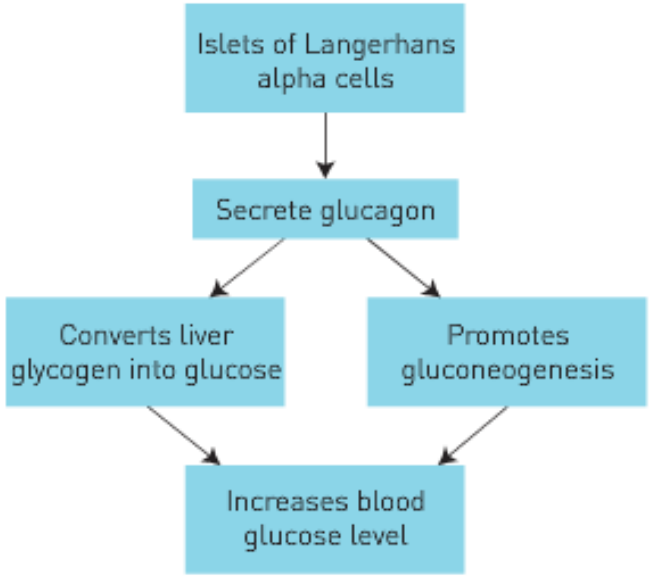
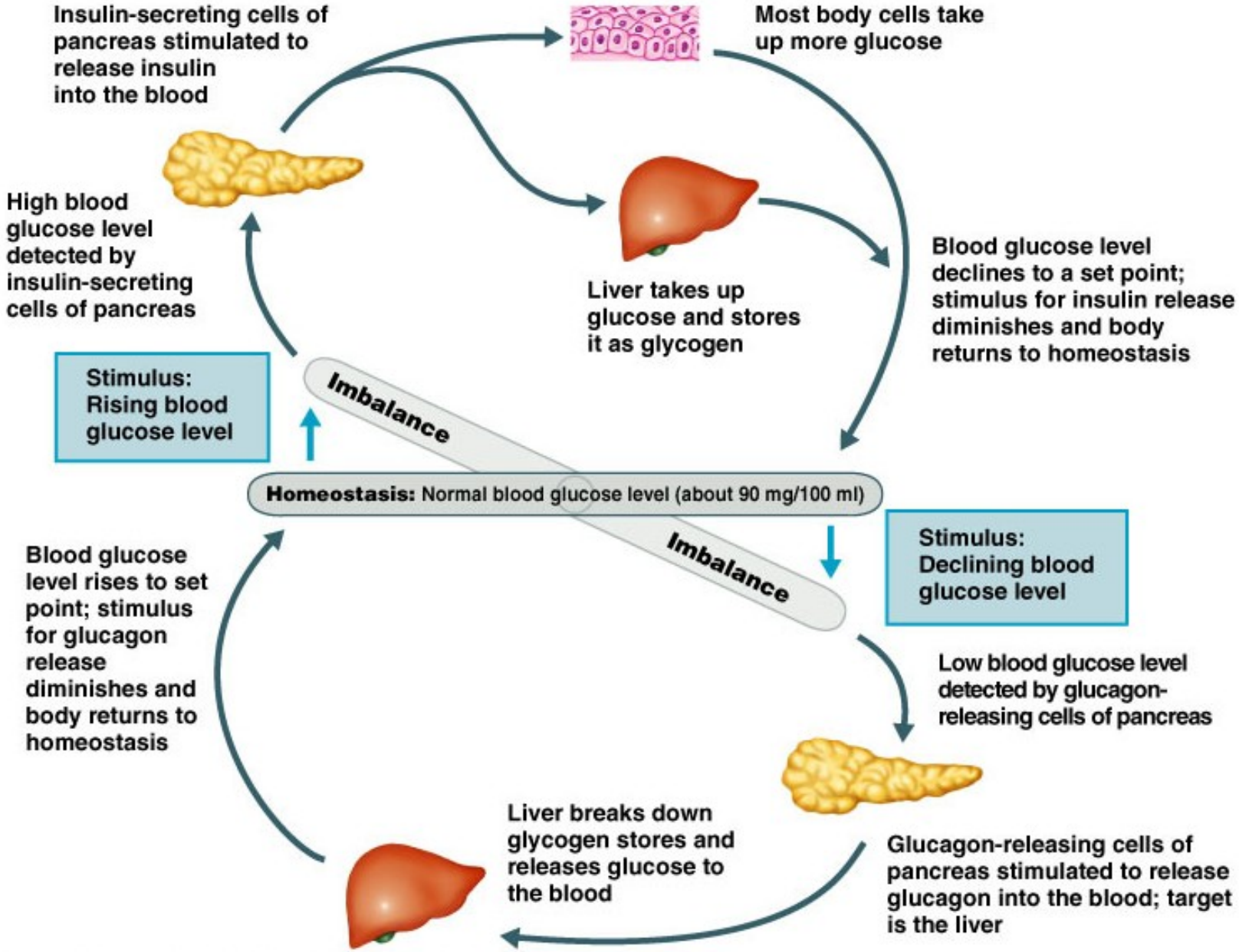


Figure 8.8 Effect of glucagon on blood glucose levels

Table 8.1 Terms relating to glucose metabolism

Term	Meaning
Glycogenesis	Formation of glycogen from other carbohydrates, especially glucose (in Greek, *genesis* means 'origin' or 'creation')
Glycogenolysis	Breakdown of glycogen to glucose (in Greek, *lysis* means 'to separate or break down')
Gluconeogenesis	Conversion of fats or proteins into glucose (in Greek, *neo* means 'new')



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Effect of the Adrenal Glands on Blood Glucose Levels

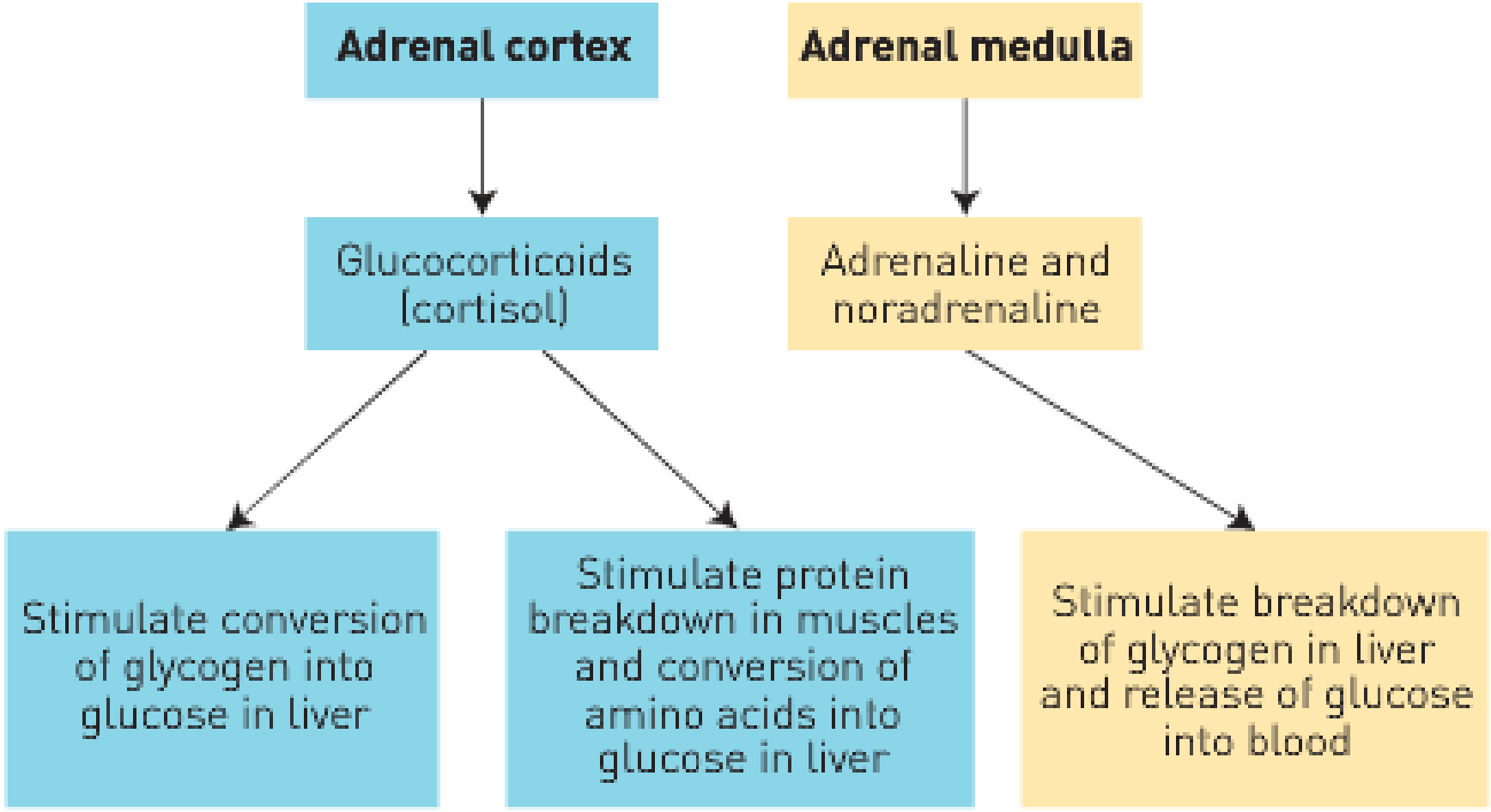


Figure 8.10 Effect of adrenal hormones on blood glucose levels

Homeostasis when blood glucose rises

Stimulus: Increased blood glucose due to: digestive processes/increased thyroid hormone/increased cortisol/increased adrenalin or nAdr release

Receptor: Beta cells in Pancreatic Islets

Message: Signalling within pancreas



Control Centre / Modulator: Beta cells in Pancreatic Islets

Message: Insulin release

Effector: Body cells | Liver

Response: Glucose enters cells to be used for cellular respiration | Liver takes up excess glucose and converts it to glycogen for storage (glycogenesis)

Feedback: Negative Feedback: Blood glucose decreases (due to uptake by cells and liver). Stimulus no longer present, cycle stops

Homeostasis when blood glucose rises above set point

1: Stimulus
Rising blood glucose

2: Receptor
Rising blood glucose detected by beta cells in pancreatic islets

Messenger:
Signalling within pancreatic islets

3: Modulator/Control Centre
Beta cells in pancreatic islets

Initiate process to lower blood glucose

Messenger:
Insulin

4: Effectors
Liver
Body cells
Fat Cells

5: Response
Liver:
- Glycogenesis
- Lipogenesis once glycogen stores are full

Body Cells:
- Absorb glucose

Fat Cells
- convert excess glucose to fat via lipogenesis

remove glucose from bloodstream

6: Feedback
Negative feedback:
Blood glucose falls back to set point
Stimulus Ceases
Cycle Stops

Homeostasis when blood glucose falls

Stimulus — Decreased blood glucose due to long time since last meal, exercise.

Receptor — Alpha cells in Pancreatic Islets

Message — Signalling within pancreas

Control Centre / Modulator — Alpha cells in Pancreatic Islets

Message — Glucagon release



Effector — Liver

Response — Glycogenolysis—liver breaks down stored glycogen into glucose and releases it to blood

Gluconeogenesis—liver converts protein and fats to glucose and releases it into blood.

Feedback — Negative Feedback: Blood glucose increases
Stimulus no longer present, cycle stops

Homeostasis when blood glucose falls below set point

1: Stimulus
Falling blood glucose

2: Receptor
Falling blood glucose detected by
a) alpha cells in pancreatic islets
b) and c) hypothalamus

Messenger:
a) Signalling within pancreatic islets
b) CRH->ACTH-> adrenal cortex
c) Sympathetic NS to adrenal medulla

3: Modulator/Control Centre
a) alpha cells in pancreatic islets
b) adrenal cortex
c) adrenal medulla
Initiate process to raise blood glucose

Messenger:
a) Glucagon
b) Cortisol
a) Adrenalin/Noradrenalin

4: Effectors
Liver
Body cells

5: Response
Liver:
- Glycogenolysis
- Gluconeogenesis
- Lipolysis

Body Cells:
- Lipolysis
- Glycogenolysis (muscles)

release glucose into bloodstream

6: Feedback
Negative feedback:
Blood glucose rises back to set point
Stimulus Ceases
Cycle Stops